Anuj Jitendra Diwan

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Research Interests

I am interested in building high-quality speech generation systems in a prosody-aware, multilingual manner.

Education

University of Texas at Austin — **Austin, TX, USA**
PhD in Computer Science — *Aug 2021 - Present*
Co-advised by Prof. David Harwath and Prof. Eunsol Choi. GPA 4.00/4.00

Indian Institute of Technology Bombay — **Mumbai, India**
B.Tech in Computer Science and Engineering with Honors — *Jul 2017 - Aug 2021*
Minor in Applied Statistics and Informatics
B.Tech CPI: 9.75/10 , Minor CPI: 10/10

Relevant Research Experience

PhD Student, UT Austin Computer Science — **Austin, TX, USA**
Advisors: Prof. David Harwath, Prof. Eunsol Choi — *Aug 2021 - Present*
- Ongoing: Prosody-preserving speech translation and style-controllable multilingual speech generation.
- Low-resource textless speech-to-speech translation using neural codec LMs (**Preprint**).
- Failures of multimodal models on visuolinguistic compositionality benchmarks (EMNLP 2022 **Paper**).
- Fine-grained computational efficiency efficiency analysis of multimodal models (ACL 2023 **Paper**).

Student Researcher, Google DeepMind — **Mountain View, CA, USA**
Advisors: Yu Zhang, Ankur Bapna — *May - Dec 2023*
- Improving TTS and speech translation robustness and generalizability via chain-of-thought prompting for speech LLMs.

Research Intern (AI), Meta AI (FAIR) — **Seattle, WA, USA**
Advisors: Abdelrahman Mohamed, Wei-Ning Hsu, Ching-Feng Yeh — *May - Dec 2022*
Continual Learning for On-Device Speech Recognition using Disentangled Conformers (ICASSP 2023) **(Paper)**
- Benchmarks and models for parameter-efficient continual learning for speaker-specific domain adaptation.

Undergraduate Researcher, IIT Bombay Computer Science — **Mumbai, India**
Advisors: Prof. Preethi Jyothi, Prof. Sunita Sarawagi — *Jan 2020 - Apr 2021*
- Co-organized the Multilingual and Code-Switching ASR Challenge, releasing labelled speech data in 7 Indian languages, strong Kaldi and ESPNet baselines, and an evaluation harness (Interspeech 2021 **Paper**).
- Transliteration-based pretraining for high-to-low resource language transfer (Interspeech 2021 **Paper**).
- Phoneme-inspired reduce-and-reconstruct learning technique for low-resource ASR (Interspeech 2021 **Paper**).
- Speech-grounded text transliteration system that only requires monolingual speech training data (**Report**).

Publications

(* indicates equal contribution)

1. **Anuj Diwan**, Anirudh Srinivasan, David Harwath, Eunsol Choi (2023). *"Unit-based Speech-to-Speech Translation Without Parallel Data"*, ArXiV Preprint. **[paper]**.
2. **Anuj Diwan**, Eunsol Choi, David Harwath (2023). *"When to Use Efficient Self Attention? Profiling Text, Speech and Image Transformer Variants"*, **ACL 2023**. **[paper][poster]**
3. **Anuj Diwan**, Ching-Feng Yeh, Wei-Ning Hsu, Paden Tomasello, Eunsol Choi, David Harwath, Abdelrahman Mohamed. *"Continual Learning for On-Device Speech Recognition using Disentangled Conformers"*, **ICASSP 2023**. **[paper][poster]**
4. **Anuj Diwan***, Layne Berry*, Eunsol Choi, David Harwath, Kyle Mahowald. *"Why is Winoground Hard? Investigating Failures in Visuolinguistic Compositionality"*, **EMNLP 2022**. **[paper][code][slides]**
5. **Anuj Diwan***, Puyuan Peng*, Raymond J. Mooney. *"Zero-shot Video Moment Retrieval With Off-the-Shelf Models"*, **TL4NLP Workshop @ NeurIPS 2022**. **[paper][poster]**
6. **Anuj Diwan**, Preethi Jyothi. *"Reduce and Reconstruct: ASR for Low-Resource Phonetic Languages"*, **INTERSPEECH 2021**. 🏆 Shortlisted for the **Best Student Paper Award**. **[paper][slides]**

7. **Anuj Diwan**, Rakesh Vaideeswaran, Sanket Shah, Ankita Singh et. al. *"MUCS 2021: Multilingual and code-switching ASR challenges for low resource Indian languages"*, **INTERSPEECH 2021**. **[paper][code]**
8. Shreya Khare*, Ashish Mittal*, **Anuj Diwan***, Sunita Sarawagi, Preethi Jyothi, Samarth Bharadwaj. *"Low Resource ASR: The surprising effectiveness of High Resource Transliteration"*, **INTERSPEECH 2021**. **[paper][slides]**

Internships

Student Researcher, Google DeepMind — **Mountain View, CA, USA**
Advisors: Yu Zhang, Ankur Bapna — *May - Dec 2023*
Chain-of-Thought Prompting for Speech-Text LLMs

Research Intern (AI), Meta AI (FAIR) — **Seattle, WA, USA**
Advisors: Abdelrahman Mohamed, Wei-Ning Hsu, Ching-Feng Yeh — *May - Dec 2022*
Continual Learning for On-Device Speech Recognition using Disentangled Conformers

Research Intern, Big Data Experience Lab, Adobe Research — **Bangalore, India**
Advisors: Sunav Choudhary, Subrata Mitra — *Apr - Jul 2020*
Learning to Learn with Interruptions

Research Intern, INMA, ICTEAM, UCLouvain — **Louvain-la-Neuve, Belgium**
Advisor: Prof. Pierre-Antoine Absil — *May - Jul 2019*
Graph-regularized Matrix Completion using Riemannian manifolds **[report][code]**

Professional Services

Reviewer: ACL 2024 (ACL Rolling Review), ICASSP 2024, EMNLP 2023, IEEE JSTSP.
Teaching Assistant: Speech Recognition, Logic for CS, Computer Programming.

Awards

- Shortlisted for the ISCA Best Student Paper Award at Interspeech 2021 for 'Reduce and Reconstruct'.
- Awarded the Excellence in Research Award 2021 by the IIT Bombay CS Department.
- Awarded the Excellence in Teaching Assistantship Award 2021 by the IIT Bombay CS Department.
- Awarded the IIT Bombay Undergraduate Research Award (URA 01), Spring 2020.
- Awarded 7 AP grades for outstanding performance (top 1%) in 7 courses.
- Awarded the Institute Academic Prize for Academic Excellence by IIT Bombay for 2017-18.
- Among top 35 in India selected for the International Mathematical Olympiad Training Camp in 2016.
- Achieved All India Rank 118 in JEE Advanced 2017 and All India Rank 197 in JEE Mains 2017.

Programming Skills

Programming Languages & Tools: Python, C/C++, Java, Bash, MATLAB, Javascript, Git, LaTeX, Beamer.
AI/ML Libraries: Pytorch, Tensorflow, Numpy, Fairseq, ESPNet, Kaldi, OpenFST, Huggingface Transformers.

Other Research Experience

Stem2Morph: Low-Resource Morphological Inflection **[slides][code]**
Course Project (Natural Language Processing). Advisor: Prof. Pushpak Bhattacharya — *Jan - May 2021*
- Reimplemented the 'Pushing the Limits of Low-Resource Morphological Inflection' paper's original Dynet implementation in Pytorch. Found that transfer learning using related languages helps. Built a user-friendly demo.

Negative Interference in Multilingual Models and Code-Switching **[slides]**
Course Project (Advanced Machine Learning). Advisor: Prof. Sunita Sarawagi — *Jan - May 2021*
- Showed that pretraining a bilingual Hindi-English model results in worse Hindi-English code-switching performance compared to monolingual pretraining, indicating negative interference and proposed potential fixes.

Anuvaadya: Instrumental Music Translation **[report][code]**
Course Project (Automatic Speech Recognition). Advisor: Prof. Preethi Jyothi — *Aug - Nov 2019*
- Implemented the 'A Universal Music Translation Network' paper that uses a CNN WaveNet Autoencoder and tested it for Indian instruments. Extended the paper's idea to arbitrary length sequence data by implementing LSTM RNN autoencoders in Pytorch inspired by the novel elements of the paper (attention and a domain confusion network).

Learning to Learn with Interruptions

Research Intern. Advisors: Sunav Choudhary, Subrata Mitra, Adobe Research *Apr - Jul 2020*

- Designed a novel reinforcement learning-based scheduler that can run ML jobs on interruptible AWS cloud-based VMs.

Graph-regularized Matrix Completion using Riemannian manifolds **[report][code]**

Research Intern. Advisor: Prof. Pierre-Antoine Absil, UCLouvain *May - Jul 2019*

- Implemented Riemannian optimization for matrix completion using graph-regularization in MATLAB and LAPACK.